



1.14

Function Model Construction and Application

Student Notes and Guided Practice

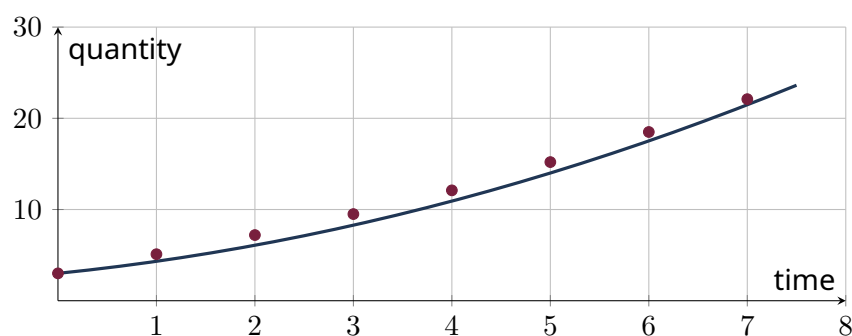
AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Construct function models from formulas, data, transformations, and constraints.
- Solve for model parameters and interpret them in context.
- Use models for interpolation, cautious extrapolation, and inverse prediction.

Essential Question

How can conditions determine a model and how should its predictions be interpreted?



Concept Framework

Key Idea

Choose a model family before solving for parameters.

Key Idea

Each independent condition usually supplies one equation for one unknown parameter.

Key Idea

Parameters should be interpreted with units and contextual meaning.

Key Idea

Model output is an estimate whose reliability depends on assumptions and distance from observed data.

Formula Map**Formula and Structure**

$$\text{linear: } f(x) = mx + b$$

Formula and Structure

$$\text{quadratic: } f(x) = a(x - h)^2 + k$$

Formula and Structure

$$\text{exponential: } f(x) = ab^x$$

Worked Examples**Worked Example 1**

Problem. Construct a linear model through $(2, 11)$ and $(7, 31)$.

Solution. Slope $m = 4$. Using $11 = 8 + b$, $b = 3$. Model: $f(x) = 4x + 3$.

Worked Example 2

Problem. A quadratic has vertex $(3, -2)$ and passes through $(5, 6)$. Find it.

Solution. Use $f(x) = a(x - 3)^2 - 2$. Then $6 = 4a - 2$, so $a = 2$.

Worked Example 3

Problem. An exponential model has $f(0) = 12$ and $f(4) = 30$. Find the four-unit growth factor.

Solution. The factor over four units is $30/12 = 2.5$. The one-unit factor is $2.5^{1/4}$.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Construct the model: linear through $(1,7)$, $(5,23)$.

2. Construct the model: quadratic vertex $(2,-1)$, through $(4,11)$.

3. Construct the model: exponential with $f(0)=8$, $f(2)=18$.

4. Construct the model: direct variation with $y=35$ when $x=5$.

5. Construct the model: inverse variation with $y=4$ when $x=6$.

6. A quadratic has zeros -2 and 5 and passes through $(0,-20)$. Find it.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.